Classification Report - RandomForestClassifier

Validation: k_folds · generated by ml_pipe

1. Executive Summary

model	RandomForestClassifier
validation	k_folds
accuracy	0.9492
precision	0.9494
recall	0.9492
f1	0.9493
fit + predict time	0.898 s

2. Experiment Configuration

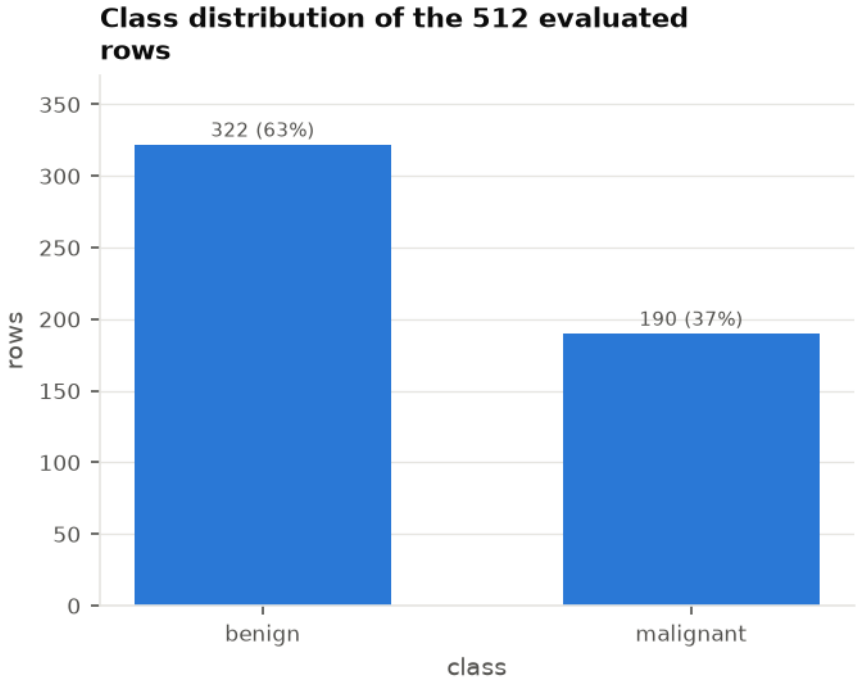
bootstrap	True
ccp_alpha	0.0
class_weight	None
criterion	'gini'
max_depth	8
max_features	'sqrt'
max_leaf_nodes	None
max_samples	None
min_impurity_decrease	0.0
min_samples_leaf	1
min_samples_split	2
min_weight_fraction_leaf	0.0
monotonic_cst	None
n_estimators	100
n_jobs	None
oob_score	False
random_state	0
verbose	0
warm_start	False

3. Classification Analysis

	precision	recall	f1-score	support
0	0.96	0.96	0.96	322
1	0.93	0.94	0.93	190
accuracy			0.95	512
macro avg	0.94	0.95	0.95	512
weighted avg	0.95	0.95	0.95	512

Class distribution

id: clf_class_distribution

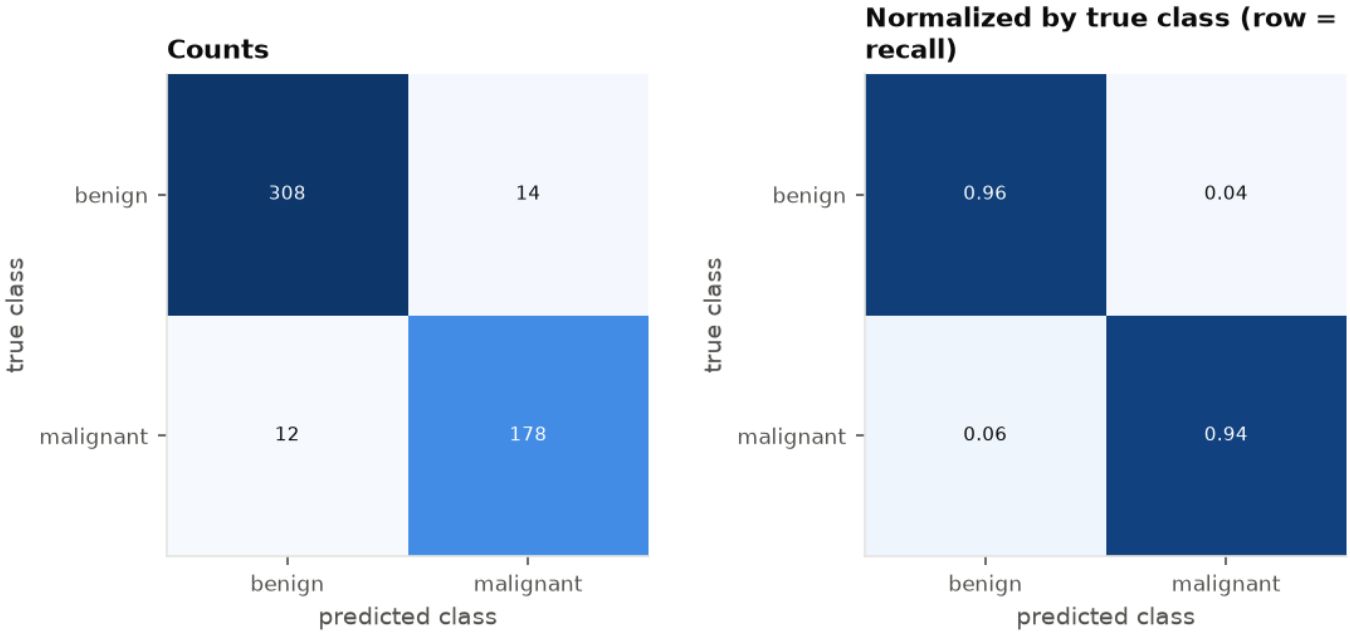


What it shows. Number of rows per class among the rows the scores were computed on (all rows for k-fold / leave-one-out, the test rows for a train / test split). Strong imbalance makes accuracy misleading - read the per-class and precision-recall plots.

classes	2
rows	512
largest / smallest class	1.69

Confusion matrix - RandomForestClassifier

id: clf_confusion:RandomForestClassifier · model(s): RandomForestClassifier

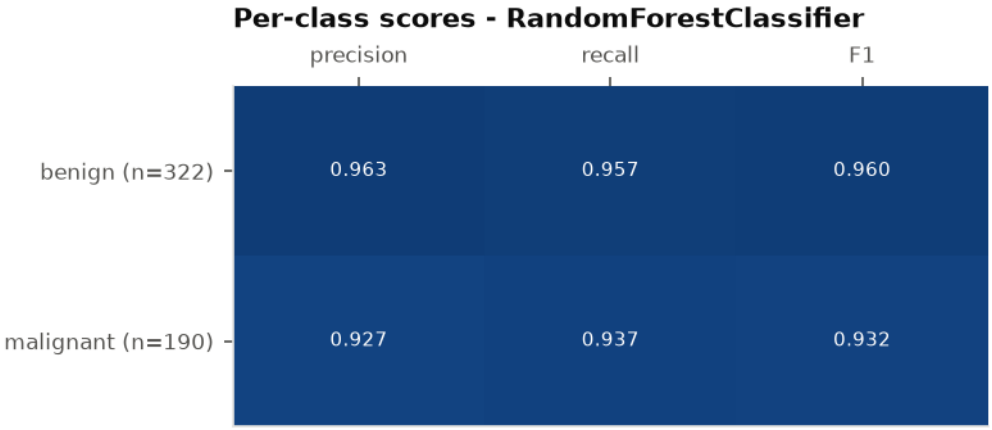


What it shows. Rows are the true classes, columns the predictions. Left: counts. Right: each row divided by its total, so the diagonal is per-class recall and off-diagonal cells show which classes are mistaken for which - independent of class size.

accuracy	0.9492
lowest recall	malignant (0.937)
rows	512

Per-class precision, recall and F1

id: clf_per_class · model(s): RandomForestClassifier

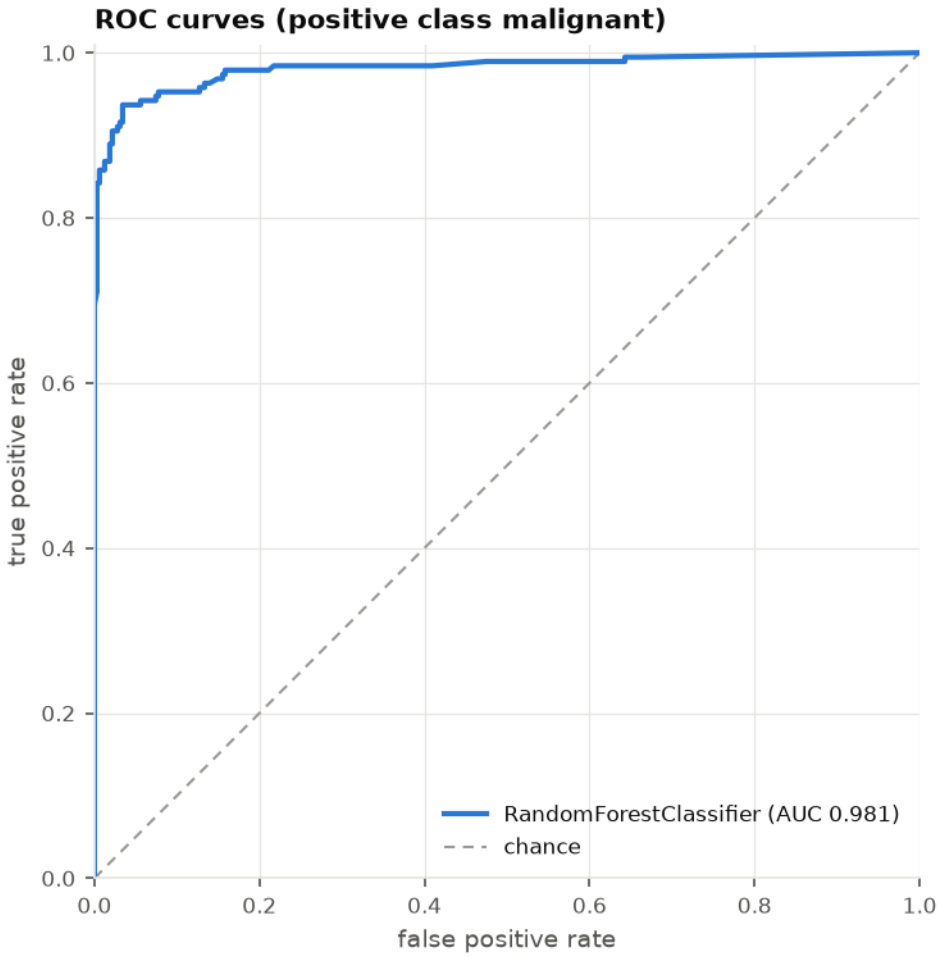


What it shows. Precision, recall and F1 of every class for the best model, with each class's support (n). The weighted averages in the leaderboard can hide a class the model almost never gets right; scores of small classes are noisy.

macro F1	0.9457
weighted F1	0.9493
worst class (F1)	malignant (0.932)

ROC curves

id: clf_roc · model(s): RandomForestClassifier

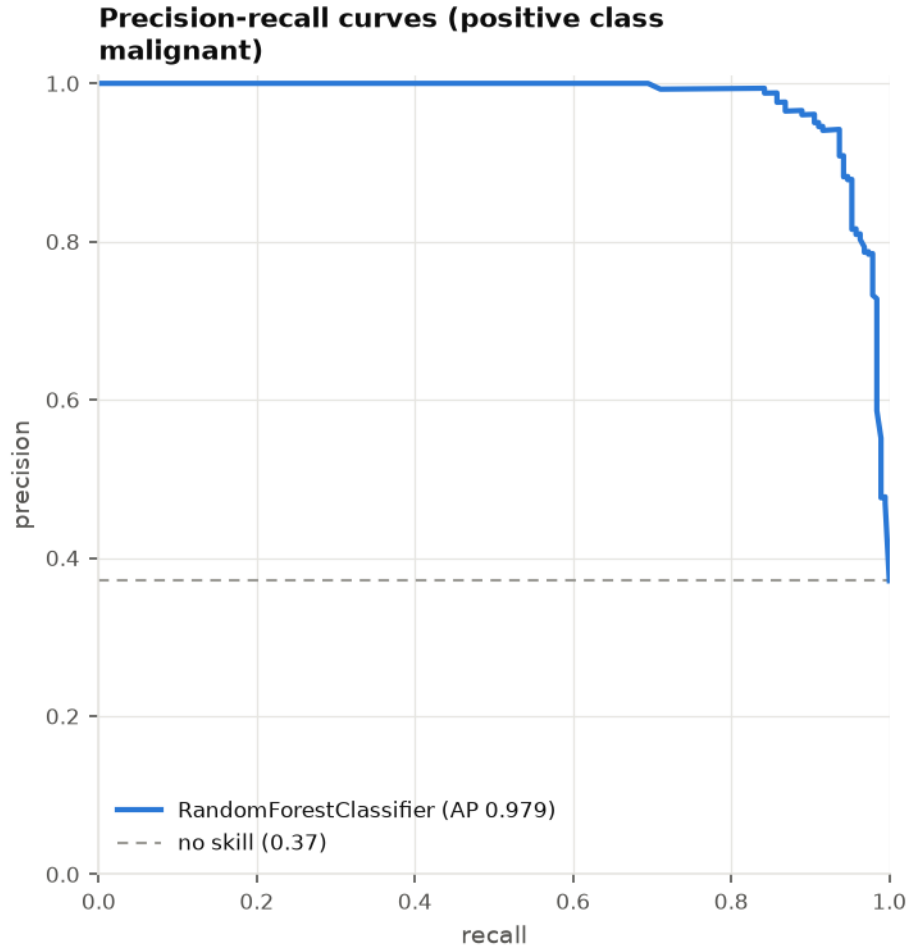


What it shows. True vs false positive rate over every decision threshold, for each model's best configuration (multiclass: micro-average over one-vs-rest classes). AUC is the probability that a random positive is ranked above a random negative; 0.5 is chance. Can look optimistic when positives are rare - see the precision-recall curves.

RandomForestClassifier ROC AUC	0.9815

Precision-recall curves

id: clf_pr · model(s): RandomForestClassifier

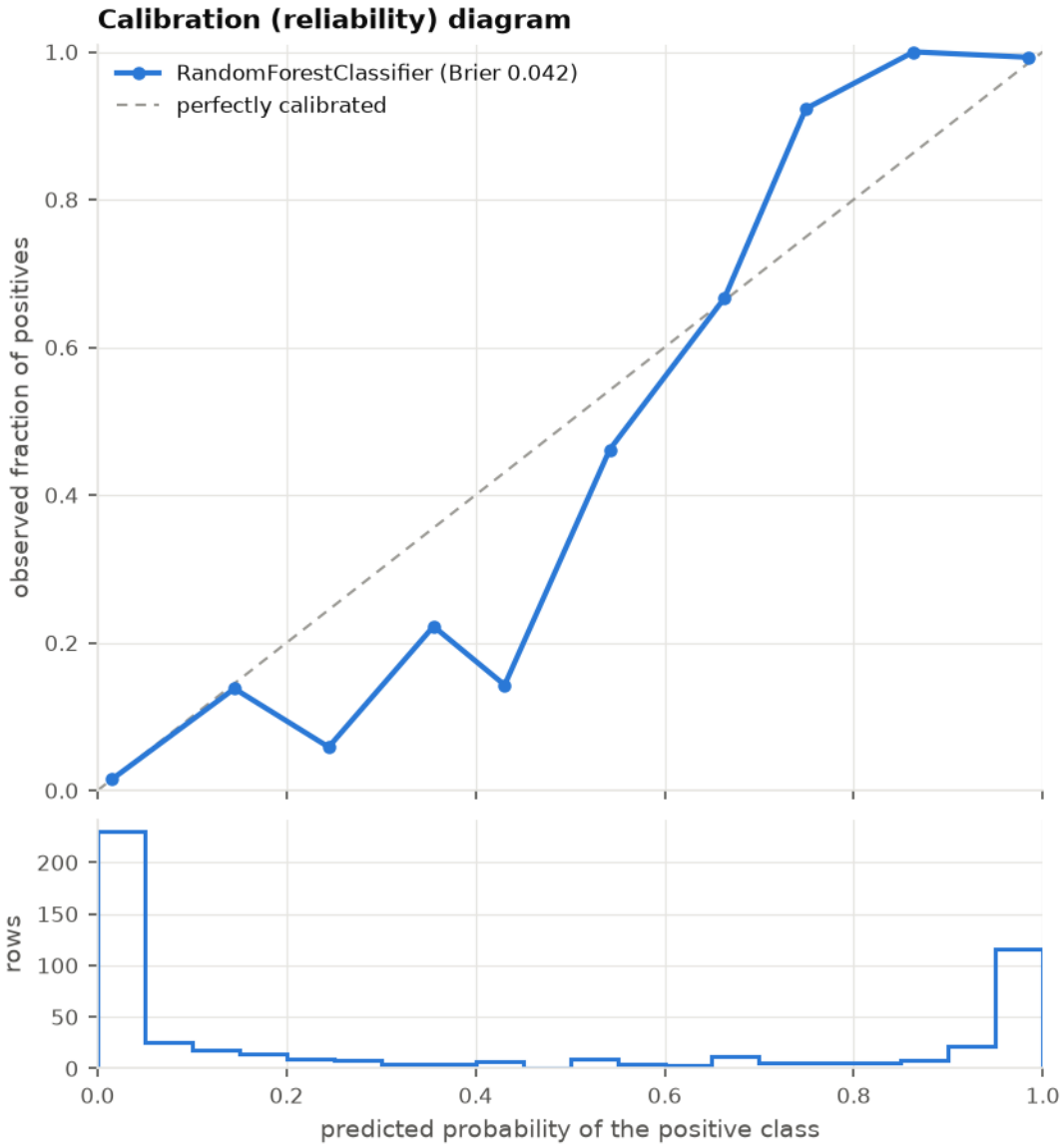


What it shows. Precision against recall over every threshold (multiclass: micro-average). Average precision (AP) summarises the curve; the dashed line is a classifier with no skill (the positive rate). More informative than ROC when the positive class is rare.

RandomForestClassifier average precision	0.9785
no-skill precision (prevalence)	0.3711

Calibration

id: clf_calibration · model(s): RandomForestClassifier



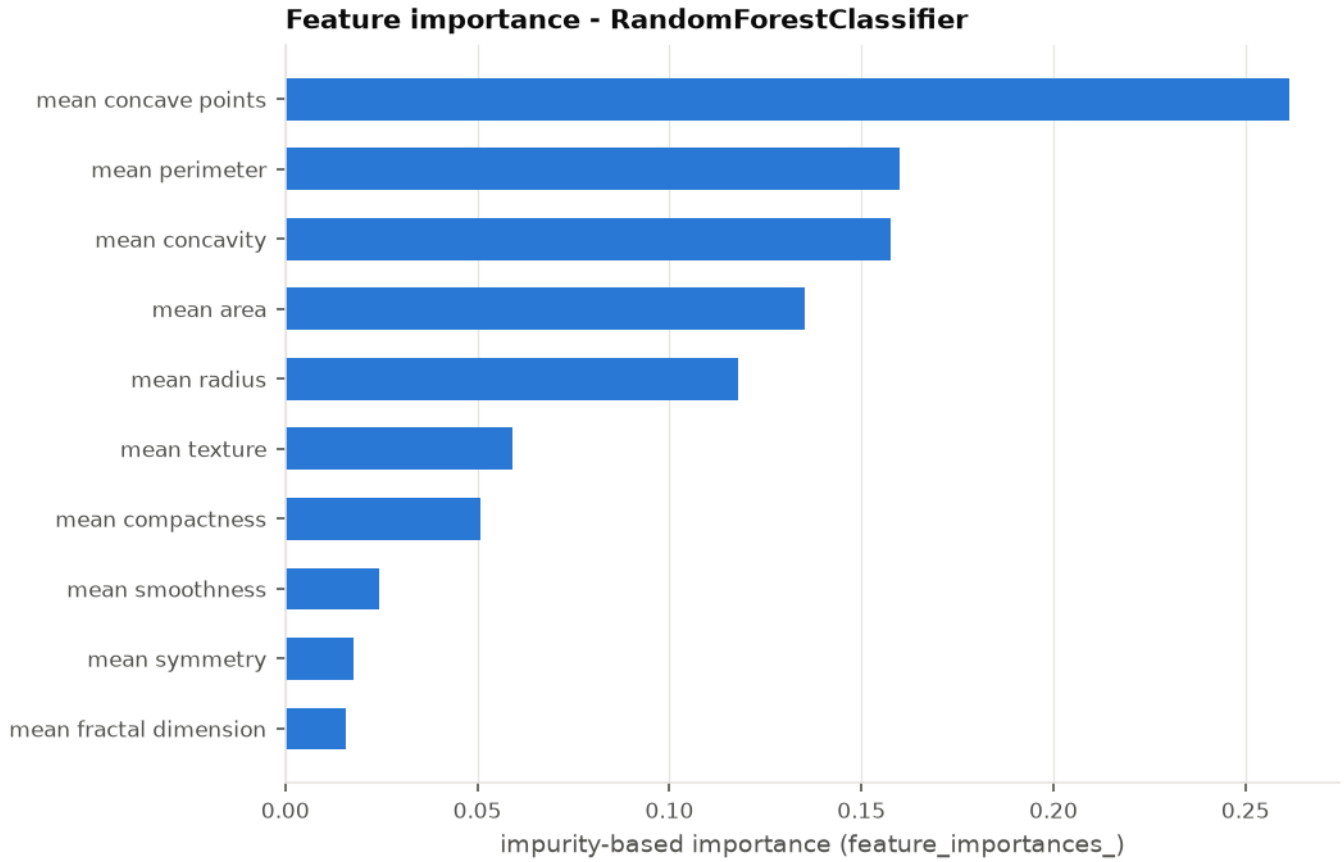
What it shows. Whether predicted probabilities can be read as probabilities: in each probability bin, the observed share of positives against the mean prediction (diagonal = perfect). Below the diagonal the model is over-confident. The histogram shows how many rows fall in each bin; the Brier score (lower is better) combines calibration and discrimination.

RandomForestClassifier Brier score	0.0418

4. Feature Analysis

Feature importance - RandomForestClassifier

id: feature_importance:RandomForestClassifier · model(s): RandomForestClassifier



What it shows. The importance the fitted model itself assigns to each input: feature_importances_ for tree ensembles, the absolute coefficient for linear models. Shows what the model relies on, not causal effects; correlated features share importance. For a model-agnostic, held-out measure use FeatureResearch.permutation_importance().

Impurity importance favours continuous / high-cardinality features and is computed on training data.

mean concave points	0.2613
mean perimeter	0.1599
mean concavity	0.1578
mean area	0.1352
mean radius	0.1180
mean texture	0.0591
mean compactness	0.0508
mean smoothness	0.0243
mean symmetry	0.0178
mean fractal dimension	0.0159

5. Visualization Index

Every visualization attempted for this report: 7 generated, 2 skipped, 0 failed.

#	visualization	section	status	time (s)	detail
1	Class distribution	Classification Analysis	SUCCESS	0.07	included in the report
2	Confusion matrix - RandomForestClassifier	Classification Analysis	SUCCESS	0.10	included in the report
3	Per-class precision, recall and F1	Classification Analysis	SUCCESS	0.06	included in the report
4	ROC curves	Classification Analysis	SUCCESS	0.14	included in the report
5	ROC per class	Classification Analysis	SKIPPED	0.00	binary target: the single curve is in the model comparison plot
6	Precision-recall curves	Classification Analysis	SUCCESS	0.11	included in the report
7	Precision-recall per class	Classification Analysis	SKIPPED	0.00	binary target: the single curve is in the model comparison plot
8	Calibration	Classification Analysis	SUCCESS	0.16	included in the report
9	Feature importance - RandomForestClassifier	Feature Analysis	SUCCESS	0.12	included in the report

6. Errors, Warnings and Missing Visualizations

Skipped visualizations (2)

Not applicable to the data or results available - nothing was estimated to fill the gap.

ROC per class: binary target: the single curve is in the model comparison plot

Precision-recall per class: binary target: the single curve is in the model comparison plot